

Discussion on “Robo-writers”

The article by Matthew Hutson (2021) examines the capabilities and limitations of large language models such as GPT-3. These systems demonstrate remarkable fluency in generating human-like text across a wide range of tasks, including writing essays, answering questions, and producing code. Their performance is largely driven by training on vast amounts of textual data—GPT-3, for instance, was trained on approximately 200 billion words. This scale enables the model to capture complex statistical patterns in language.

However, a central argument of the article is that these models do not possess genuine understanding. Instead, they operate by predicting likely word sequences based on learned correlations. As a result, their outputs can be coherent but conceptually flawed or nonsensical. The article highlights that such systems can produce incorrect statements or even harmful responses, demonstrating the gap between linguistic fluency and true comprehension.

Another important concern is bias. Because language models are trained on large, unfiltered datasets, they can reproduce and amplify societal biases present in the data. This includes generating stereotypical or offensive content, even when not explicitly prompted. The article also notes that increasing model size improves performance but does not resolve these fundamental issues, suggesting that scaling alone is not a sufficient solution.

Since the publication of the article, large language models have advanced significantly. Modern systems, such as GPT-5-level models, are substantially more capable than GPT-3, although exact parameter counts are often not publicly disclosed. While earlier models such as GPT-3 had 175 billion parameters, newer models are believed to operate at a much larger scale and incorporate architectural improvements beyond simple parameter growth. In addition, recent developments include so-called reasoning models, which are designed to improve multi-step problem solving, logical consistency, and task decomposition. These models extend beyond pure next-word prediction by integrating structured reasoning processes, often through techniques such as chain-of-thought prompting or reinforcement learning. Despite these advancements, many of the fundamental limitations identified in the article—such as lack of true understanding and sensitivity to training data—remain relevant, although they are often less visible in practice.

In conclusion, while large language models represent a significant technological advancement, their limitations—particularly the lack of understanding and the presence of bias—raise important questions about their reliability and appropriate use.

References

Hutson, M. (2021) ‘The language machines’, *Nature*, 591, pp. 22–25.

Gradient descent

The code:

```
# code credit:codebasics https://codebasics.io/coming-soon

import numpy as np

def gradient_descent(x,y):

    m_curr = b_curr = 0

    iterations = 100          #change value

    n = len(x)

    learning_rate = 0.09      #change value

    for i in range(iterations):

        y_predicted = m_curr * x + b_curr

        cost = (1/n) * sum([val**2 for val in (y-y_predicted)])

        md = -(2/n)*sum(x*(y-y_predicted))

        bd = -(2/n)*sum(y-y_predicted)

        m_curr = m_curr - learning_rate * md

        b_curr = b_curr - learning_rate * bd

        print ("m {}, b {}, cost {} iteration {}".format(m_curr,b_curr,cost, i))

x = np.array([1,2,3,4,5])

y = np.array([5,7,9,11,13])

gradient_descent(x,y)
```

For iteration = 100 and learning rate = 0.08 we have

```
m 2.0405161418256377, b 2.8536001910078013, cost 0.004120600119124239  
iteration 99
```

While For iteration = 100 and learning rate = 0.09 the values are off the charts.

```
m -514273.8757435459, b -142443.28242641973, cost 2640508310116.111  
iteration 99
```

This model is very sensitive to these parameters, especially to the learning rate.